

RealMan Robotic Arm rm_gazebo User Manual V1.6

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Revision History

No.	Date	Comment
V1.0	2024-2-19	Draft
V1.1	2024-7-8	Amend(Add gen72 related adapter files)
V1.1.1	2024-8-13	Amend(Add robotic arm model adaptation description)
V1.2	2024-9-10	Amend(Add eco63 related adapter files)
V1.3	2024-12-25	Amend(Add 63, 65, 75, ECO65 six-axis force adapter files and 63, 65, 75, ECO63, ECO65 integrated six-axis force adapter files)
V1.4	2025-4-3	Amend(Add Gen72_II adapter files)
V1.5	2025-11-13	Amend(Add RML63_III adapter files)
V1.6	2026-4-16	Amend(Add ECO62 and RX75 adapter files, update Gazebo version)

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rm_gazebo_Package_Description

rm_gazebo is mainly used for realizing the simulation function of robot arm Moveit2 planning. We build a virtual robotic arm in the simulation environment of Gazebo, and then control the virtual robot arm in Gazebo through Moveit2. This package is introduced in detail in the following aspects.

- 1.Package use.
- 2.Package architecture description.

Through the introduction of this part, it can help you:

- 1.Understand the package use.
- 2.Familiar with the file structure and function of the package.

Source code address: https://github.com/RealManRobot/ros2_rm_robot.git.

rm_gazebo_Package_Running

Control_of_the_simulation_robotic_arm

After the installation of the environment and the package, we can run the rm_gazebo package. Use the following command to launch the Gazebo virtual space and the virtual robotic arm.

```
rm@rm-desktop:~$ ros2 launch rm_gazebo gazebo_65_demo.launch.py
```

The command to start the six-axis force version is currently available for 63, 65, 75, and eco65:

```
rm@rm-desktop:~$ ros2 launch rm_gazebo gazebo_<arm_type>_6f_demo.launch.py
```

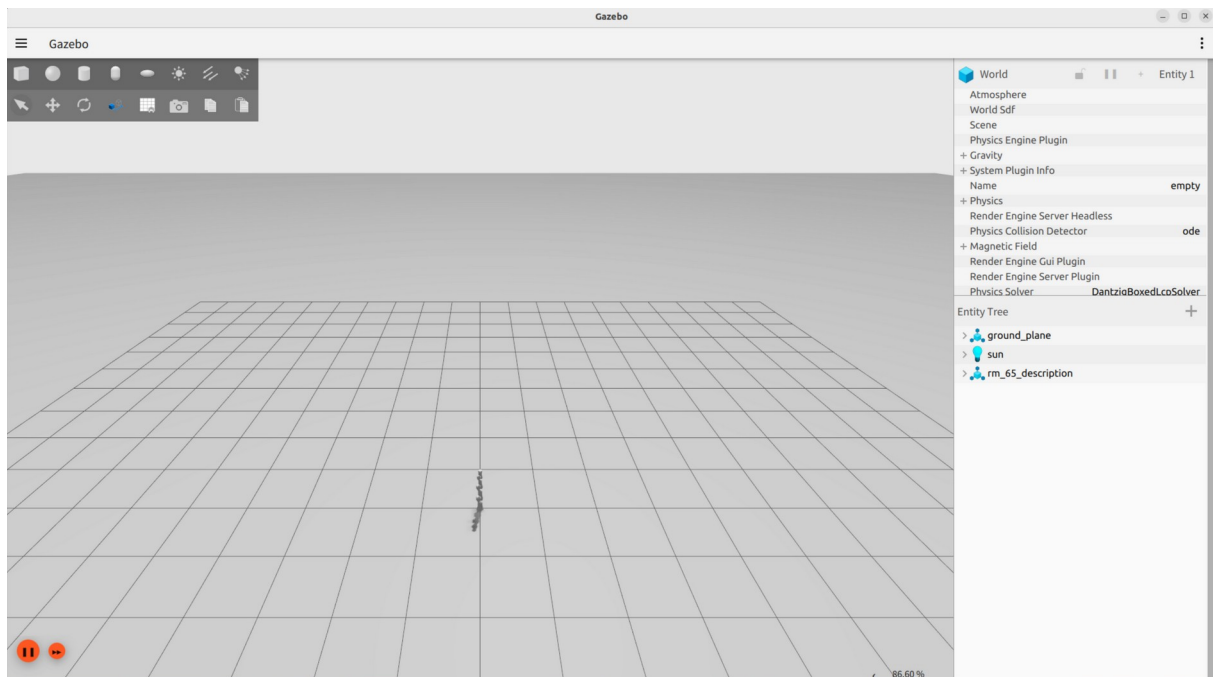
The command to start the integrated six-axis force version is currently available for 63, 63_III, 65, 75, eco63, eco65, and rx75:

```
rm@rm-desktop:~$ ros2 launch rm_gazebo gazebo_<arm_type>_6fb_demo.launch.py
```

The command to start the vision-enabled integrated six-axis force version is currently only available for rx75:

```
rm@rm-desktop:~$ ros2 launch rm_gazebo gazebo_<arm_type>_6fb_v_demo.launch.py
```

In practice, the above needs to be replaced by the actual model of the robotic arm. The available models are 65, 63, 63_III, 75, eco62, eco63, eco65, gen72, gen72_II, and rx75. For RX75, use gazebo_rx75_6fb_demo.launch.py for RX75-6FB and gazebo_rx75_6fb_v_demo.launch.py for RX75-6FB-V. The interface displays as follows after successful running.



Then, we use the following command to launch moveit2 to control the simulation robot arm in Gazebo.

```
rm@rm-desktop:~$ ros2 launch rm_<arm_type>_config gazebo_moveit_demo.launch.py
```

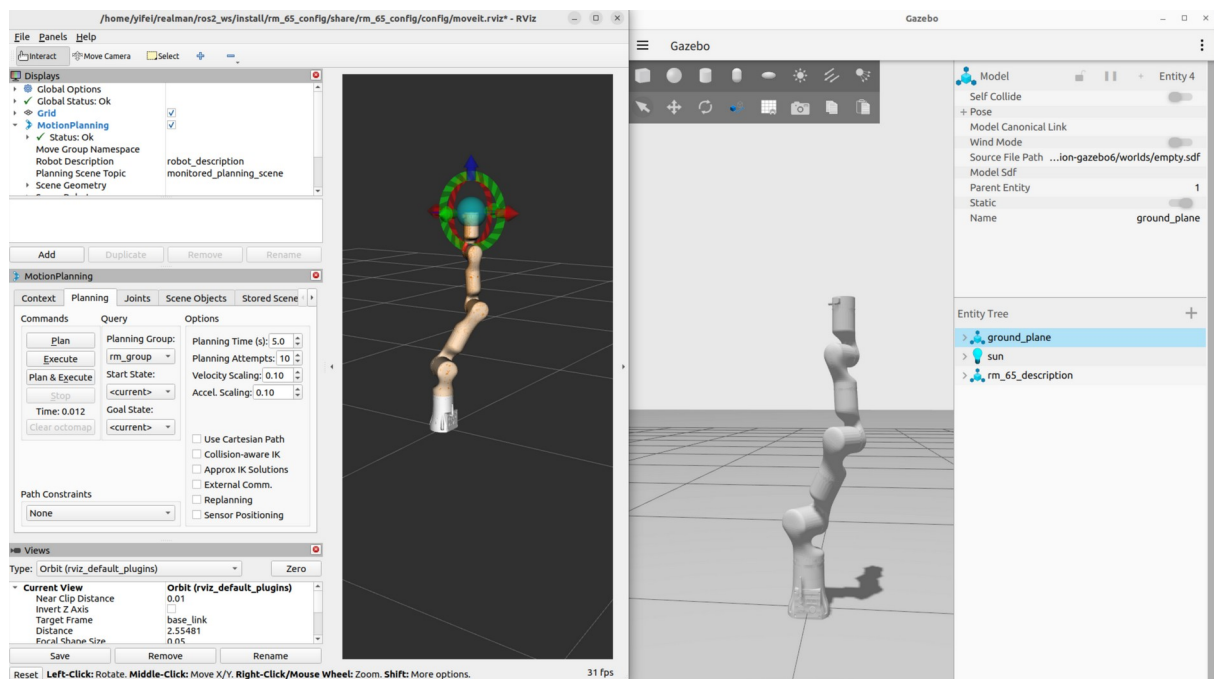
The command to start the six-axis force version is currently available for 63, 65, 75, and eco65:

```
rm@rm-desktop:~$ ros2 launch rm_<arm_type>_config gazebo_moveit_demo_6f.launch.py
```

The command to start the integrated six-axis force version is currently available for 63, 63_III, 65, 75, eco63, and eco65:

```
rm@rm-desktop:~$ ros2 launch rm_<arm_type>_config gazebo_moveit_demo_6fb.launch.py
```

In practice, the above needs to be replaced by the actual model of the robotic arm. The available models are 65, 63, 63_III, 75, eco62, eco63, eco65, gen72, and gen72_II. RX75 uses the dedicated MoveIt package `rm_rx75_config`: run `ros2 launch rm_rx75_config gazebo_moveit_demo_6fb_v.launch.py` for RX75-6FB-V and `ros2 launch rm_rx75_config gazebo_moveit_demo_6fb.launch.py` for RX75-6FB. After the control interface of rviz2 pops up, you can perform the simulation control of moveit2 and Gazebo.



rm_gazebo_Package_Architecture_Description

Overview_of_Package_Files

The current rm_gazebo package is composed of the following files.

```

├── CMakeLists.txt                # compilation rule file
├── config
│   ├── gazebo_63_6fb_description.urdf.xacro    #RML63 integrated six-axis force gazebo
├── launch file
│   ├── gazebo_63_III_description.urdf.xacro    #RML63_III integrated gazebo launch
├── file
│   ├── gazebo_65_6fb_description.urdf.xacro    #RM65 integrated six-axis force gazebo
├── launch file
│   ├── gazebo_75_6fb_description.urdf.xacro    #RM75 integrated six-axis force gazebo
├── launch file
│   ├── gazebo_eco62_description.urdf.xacro     #EC062 gazebo model description file
│   ├── gazebo_eco63_6fb_description.urdf.xacro #EC063 integrated six-axis force gazebo
├── launch file
│   ├── gazebo_eco65_6fb_description.urdf.xacro #EC065 integrated six-axis force gazebo
├── launch file
│   ├── gazebo_63_description.urdf.xacro        #63gazebo model description file
│   ├── gazebo_65_description.urdf.xacro        #65gazebo model description file
│   ├── gazebo_75_description.urdf.xacro        #75gazebo model description file
│   ├── gazebo_eco65_description.urdf.xacro     #eco65gazebo model description file
│   ├── gazebo_eco63_description.urdf.xacro     #eco63gazebo model description file
│   ├── gazebo_gen72_II_description.urdf.xacro  #gen72_IIgazebo model description file
│   ├── gazebo_gen72_description.urdf.xacro     #gen72gazebo model description file
│   ├── gazebo_rx75_6fb_description.urdf.xacro  #RX75-6FB dual-arm gazebo model
├── description file
│   ├── gazebo_rx75_6fb_v_description.urdf.xacro #RX75-6FB-V dual-arm gazebo model
├── description file
│   ├── gazebo_rx75_description.urdf.xacro      #RX75 dual-arm gazebo helper file
├── doc
│   ├── rm_gazebo1.png
│   ├── rm_gazebo2.png
├── launch
│   ├── gazebo_63_6fb_demo.launch.py           #63 integrated six-axis force gazebo launch
├── file
│   ├── gazebo_63_6f_demo.launch.py            #63 six-axis force gazebo launch file
│   ├── gazebo_63_demo.launch.py               #63 gazebo launch file
│   ├── gazebo_63_III_6fb_demo.launch.py       #63_III integrated six-axis force gazebo
├── launch file
│   ├── gazebo_63_III_demo.launch.py           #63_III gazebo launch file
│   ├── gazebo_65_6fb_demo.launch.py           #RM65 integrated six-axis force gazebo
├── launch file

```

		gazebo_65_6f_demo.launch.py	#RM65 six-axis force gazebo launch file
		gazebo_65_demo.launch.py	#RM65 gazebo launch file
		gazebo_75_6fb_demo.launch.py	#RM75 integrated six-axis force gazebo
	launch file		
		gazebo_75_6f_demo.launch.py	#RM75 six-axis force gazebo launch file
		gazebo_75_demo.launch.py	#RM75 gazebo launch file
		gazebo_eco62_demo.launch.py	#EC062 gazebo launch file
		gazebo_eco63_6fb_demo.launch.py	#EC063 integrated six-axis force gazebo
	launch file		
		gazebo_eco63_demo.launch.py	#EC063 gazebo launch file
		gazebo_eco65_6fb_demo.launch.py	#EC065 integrated six-axis force gazebo
	launch file		
		gazebo_eco65_6f_demo.launch.py	#EC065 six-axis force gazebo launch file
		gazebo_eco65_demo.launch.py	#EC065 gazebo launch file
		gazebo_gen72_II_demo.launch.py	#gen72_IIgazebo launch file
		gazebo_gen72_demo.launch.py	#gen72gazebo launch file
		gazebo_rx75_6fb_demo.launch.py	#RX75-6FB dual-arm gazebo launch file
		gazebo_rx75_6fb_v_demo.launch.py	#RX75-6FB-V dual-arm gazebo launch file
		gz_demo_common.py	#shared Gazebo launch helper
	package.xml		
	README_CN.md		
	README.md		